Md Saif Ahmed

Software Engineer | Full-Stack Developer | Competitive Programmer

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Full-stack software engineer and CSE undergraduate at MIST. **Sole developer of Hasnahena**, a live interest-free finance platform (Go API, Next.js front end) serving real members since August 2026. Works across React/Next.js, Go, Node.js and FastAPI with a testing-first habit and an ICPC-level algorithmic foundation (**38th at ICPC Asia Dhaka Regional 2025**).

Live	**~180/s**	**38th**	**1700+**
solo-built platform, since Aug 2026	postings sustained in load test	ICPC Asia Dhaka Regional 2025	problems solved • CF Specialist

EDUCATION

B.Sc. in Computer Science & Engineering 2023 – 2027 (Expected)
Military Institute of Science and Technology (MIST), Dhaka | *CGPA* **3.58** / **4.00** *(up to Level-3, Term-2)*

Higher Secondary Certificate (HSC) | Govt. M. M. College, Jashore | GPA 5.00 / 5.00 2022

TECHNICAL SKILLS

Languages	C++, Go, Python, TypeScript, JavaScript, Java, Dart, SQL, Bash
Frontend	React, Next.js, Vite, Tailwind CSS, shadcn/ui, Radix UI, MUI, TanStack Query, next-intl, Flutter
Backend	Go (chi, pgx), Node.js, Express.js, FastAPI, Hono, Prisma, SQLAlchemy, REST APIs, SSE, JWT, OAuth 2.0, RBAC
Databases	PostgreSQL (Neon, Supabase), MongoDB (Mongoose), Oracle Database (PL/SQL), MySQL
Cloud & DevOps	GitHub Actions, Docker, Vercel, Render, Cloudflare, Cloudinary, Sentry, UptimeRobot
Testing	Playwright, Vitest, Go testing (unit, integration, property, load), pytest, ESLint, golangci-lint
AI / LLM	Anthropic Claude API, Vercel AI SDK, OpenAI-compatible APIs

PROJECTS

Hasnahena — Interest-Free Cooperative Finance Platform (Solo, in production) hasnahenafoundation.com | GitHub
Next.js 16, React 19, TypeScript, Go (chi, pgx), PostgreSQL (Neon), Prisma, Tailwind, next-intl, Cloudflare

- Built and operate a platform digitising a Jashore cooperative's passbook, **live with real members since Aug 2026**: deposits, interest-free loans with guarantors, donations, disputes; member, worker and admin roles.
- pnpm monorepo: Next.js web app on Vercel and a Go API on Render that owns every balance-changing write in a **single database transaction**; fully bilingual English/Bengali; vector PDF statements and receipts; SSE support chat.
- Tested in CI on GitHub Actions: **269 Go tests** incl. integration tests under `-race`, Playwright e2e on phone and desktop, 400k-iteration money property tests, and a load test sustaining **~180 postings/s** that reconciles exactly.
- **45 Architecture Decision Records**, runbooks and post-mortems; monitored with Sentry, UptimeRobot, Prometheus.

QuestBoard — Gamified Q&A Platform for STEM Students (Team) GitHub
Python, FastAPI, SQLAlchemy, PostgreSQL (Supabase), Anthropic Claude API, Codeforces API, pytest, Flutter, Render

- Students post "quests" with point bounties. **Wrote all of the current server code**: FastAPI, 7 routers, ~40 endpoints; Postgres schema with race-safe voting (UNIQUE), trigram search, RLS, and a point ledger moving points **atomically in one transaction**.
- **Anthropic Claude API** hints, rate-limited and refunded on failure; Codeforces API daily challenges and handle verification; **70 pytest tests** run in CI against a real Postgres; deployed on Render.

HallBridge — Digital Hall Management System (Team) GitHub
Next.js 16, React 19, TypeScript, MongoDB (Mongoose), Tailwind CSS, Supabase, Google OAuth 2.0

- Seat allocation, billing and notices for **500+ residents**, with student, staff and admin dashboards and a predictive meal planner that brought food wastage to near-zero.
- Wrote ~64% of the API and ~81% of the data models; **Google OAuth 2.0** sign-in with a CSRF state cookie, password reset with expiring tokens, and compound indexes with lean, projected reads.

LocalFix — Civic Engagement Platform (Team) GitHub
React (Vite), Node.js, Express.js, Oracle Database (PL/SQL), JWT, bcrypt, Helmet, Mapbox GL

- Complaint management system where citizens report civic issues (roads, waste) to authorities; role-based access control and optimized Oracle SQL queries **cut data retrieval latency by 40%**.
- Wrote ~95% of the Express API (Helmet, rate limiting, validation, JWT) and the Oracle schema: **14 triggers, 3 stored procedures, 3 views, 27 indexes**; plus a large share of the React client.

MCC Website — MIST Computer Club Platform (Team, in production) GitHub
Next.js, Bun, Hono, PostgreSQL, JWT, Supabase, Docker Compose, GitHub Actions

- The club's live platform (Next.js, Bun/Hono API, Discord bot; Docker Compose). Built the **alumni directory end to end**: public listing, JWT-protected admin CRUD, Postgres migrations, Next.js pages; landing-page admin.
- Hardened contest-report rooms: input validation, idempotent adds and a **broken access control fix**, so read endpoints require an admin instead of any signed-in user.

GyanTori — Peer-to-Peer Book Sharing (Team) GitHub

Java, JavaFX (FXML), Maven, MongoDB

- Desktop virtual library for **500+ students** to lend, borrow and catalog books, with a MongoDB schema for inventory relationships and transaction history.

BruteBuddy — Automated Security Testing Tool GitHub

Python, requests, ThreadPoolExecutor, CLI

- CLI for **automated fuzzing** and brute-force testing of login endpoints: multi-field and generated payloads, custom headers and cookies, automatic re-authentication; parallel requests run **3× faster**.

TOOLS & SCRIPTS

- **Rank-Raven** *(Python, Selenium, BeautifulSoup, ReportLab)*: Codeforces activity reporter that generates styled PDF reports.
- **upsolve_tracker** *(Apps Script, Sheets, Codeforces API)*: schedules near-miss problems, times solves, detects AC automatically.
- **Code-Library** *(C++, LaTeX, Python)*: MIST_EagleForces ICPC team notebook with a script that builds the PDF.
- **Faculty-Evaluation** *(Python, Selenium)*: automates the university's faculty evaluation form.

EXPERIENCE

Industrial Trainee, JB Connect Ltd (JBC) 2026 | 4 weeks

Industry sessions on DevOps, backend engineering and AI/ML; tasks with the company's teams

- Team AI content-generation app (FastAPI, React): set up the backend's router → service → repository layers, env-driven settings, error-to-HTTP handlers, Alembic and Docker dev setup.
- Built the `POST /api/v1/generate` endpoint with a mock AI client behind an injectable FastAPI dependency; shipped through pull requests.

COMPETITIVE PROGRAMMING

Codeforces Specialist (max 1514) | **CodeChef** 3-Star (max 1722) | **1700+** problems solved (CF, CodeChef, LeetCode, CSES, AtCoder)

ICPC Asia Dhaka Regional 2025 *(univ. teams)*	**38th**	ICPC Asia Dhaka Regional 2024 *(univ. teams)*	**117th**
ICPC Dhaka 2025 Online Preliminary	**290th** / 1842	ICPC Dhaka 2024 Online Preliminary	**90th** / 2490
DUET IUPC 2025	**16th** / 137	NSUCEC Cybernauts IUPC 2026	**16th** / 159
CUET IUPC 2025	**28th** / 131	DUET IUPC 2026	**24th** / 139
UIU IUPC 2025	**43rd** / 161	NDUB IUPC 2026	**32nd** / 85
AUST IUPC 2025	**48th** / 129	SUST CSE Carnival IUPC 2026	**44th** / 137
MU CSE Fest IUPC 2025	**49th** / 99	IUT ICT Fest IUPC 2026	**46th** / 111
Uttara University IUPC 2025	**54th** / 122	KUET IUPC 2025	**89th** / 161
BUP IAUPC 2025	**Champion** / 28	Meta Hacker Cup 2024	**Round 2**

LEADERSHIP & TEACHING

President, MIST Computer Club Jun 2026 – Present

Vice President (May 2025 – Jun 2026) | Executive Member (May 2024 – May 2025)

- Lead club strategy and mentor **15+ core members** delivering **10+ technical events a year**; direct ICPC preparation camps and workshops reaching **200+ students**.
- As Vice President, coordinated programming workshops, contest initiatives and training for junior members.

Competitive Programming Instructor, MIST Computer Club May 2024 – Present

- Mentor **50+ junior students** in algorithms and contest strategy with an **80% retention rate**; curated **20+ problem sets** and mock contests, helping teams advance in ICPC preliminary rounds.

Web Security Instructor, MIST Cyber Security Club May 2024 – Jun 2026

- Ran hands-on **OWASP Top 10** workshops with real-world exploitation scenarios; trained **30+ students** in penetration testing and secure coding. Also Associative Member (May 2024 – Jun 2025).

Executive Director, Mentor Committee | coordinated security mentoring initiatives and learning tracks Jun 2025 – Jun 2026

Hackathons: MIST Intra Hackathon 2025 | BUP CSE Fest 2026 Hackathon (onsite)

CYBERSECURITY & CTFS

Cyber Drill 2025 (WTISD Hackathon)	**5th**	Bugcrowd USA CTF 2025	**34th global**
SMP CTF 2024	**13th**	BUET CTF 2024 (final round)	**18th**
Cyber Raid 2025 (final round)	**18th**	DIU Cybercon CTF 2025 (final round)	**18th**
BCS ICT Fest CTF 2025	**18th**	BUP CTF 2025	**18th**

Hands-on practice: PortSwigger Academy **70+ labs** (10+ categories) | PicoCTF **67+** | TryHackMe **29 rooms** | OverTheWire Natas
Training: Hands-on Ethical Hacking and Security Analysis (MIST Cyber Security Club) | eShikhon Ethical Hacking Live Course